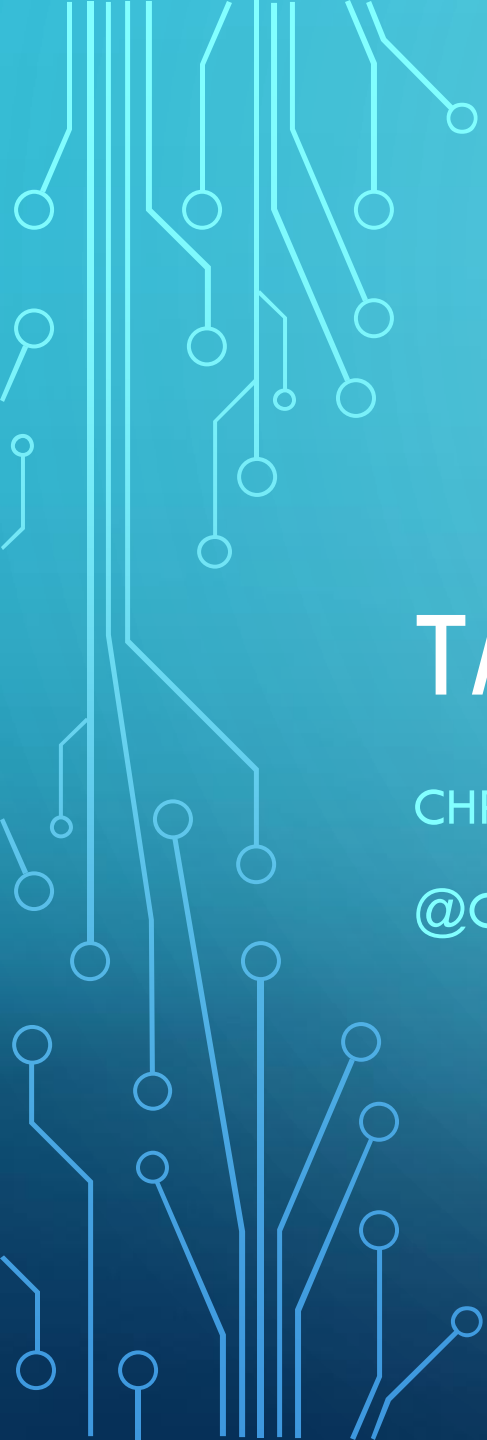


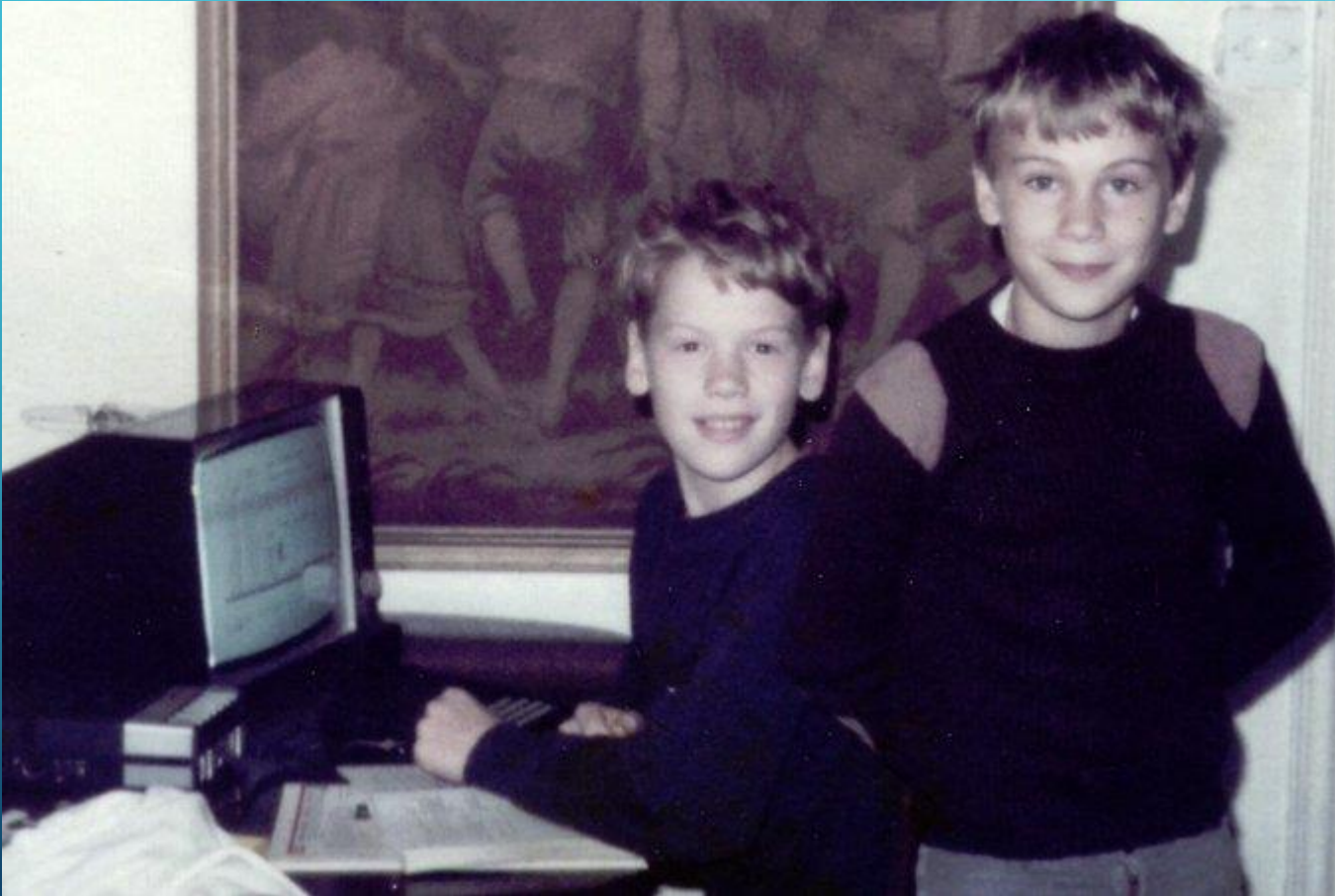


TMS9995 ON RC2014

CHRIS SWAN

@CPSWAN

RETRO ME



ZX81, Dragon 32,
Plus/4, C64

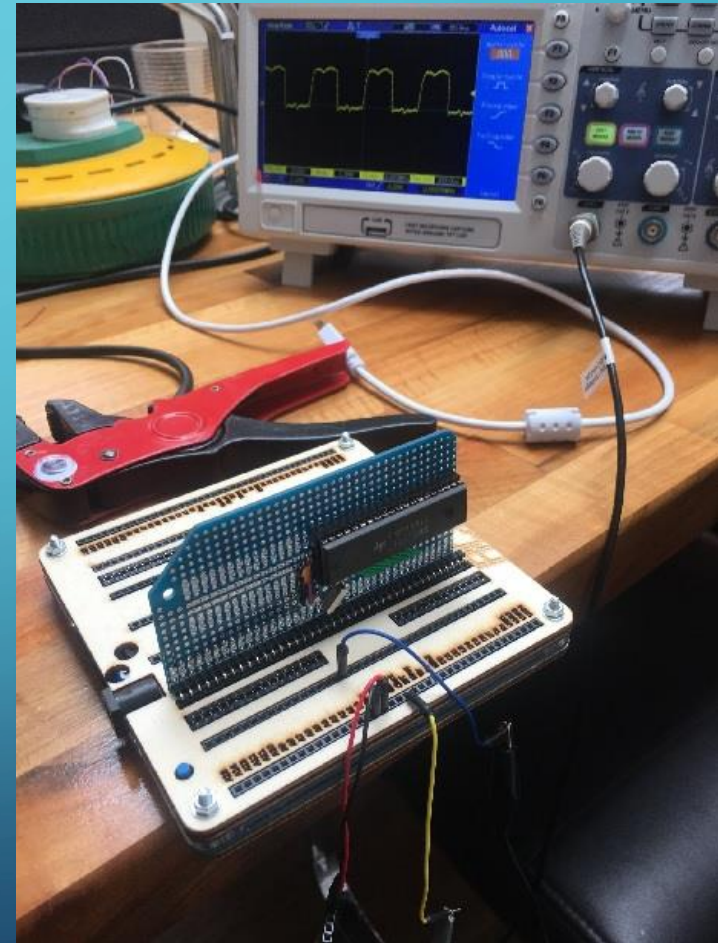
Amiga 500, PPC640
Amiga 1200

PC, PC, PC, PC...

<https://about.me/cpswan>

WHAT TO EXPECT FROM THIS TALK

- Project Inspiration
- TMS9995 origins
- A design to work from
- Putting it together
- Hello World
- Are 16bits better?



<https://hackaday.io/project/166036-tms9995-cpu-module-for-rc2014>



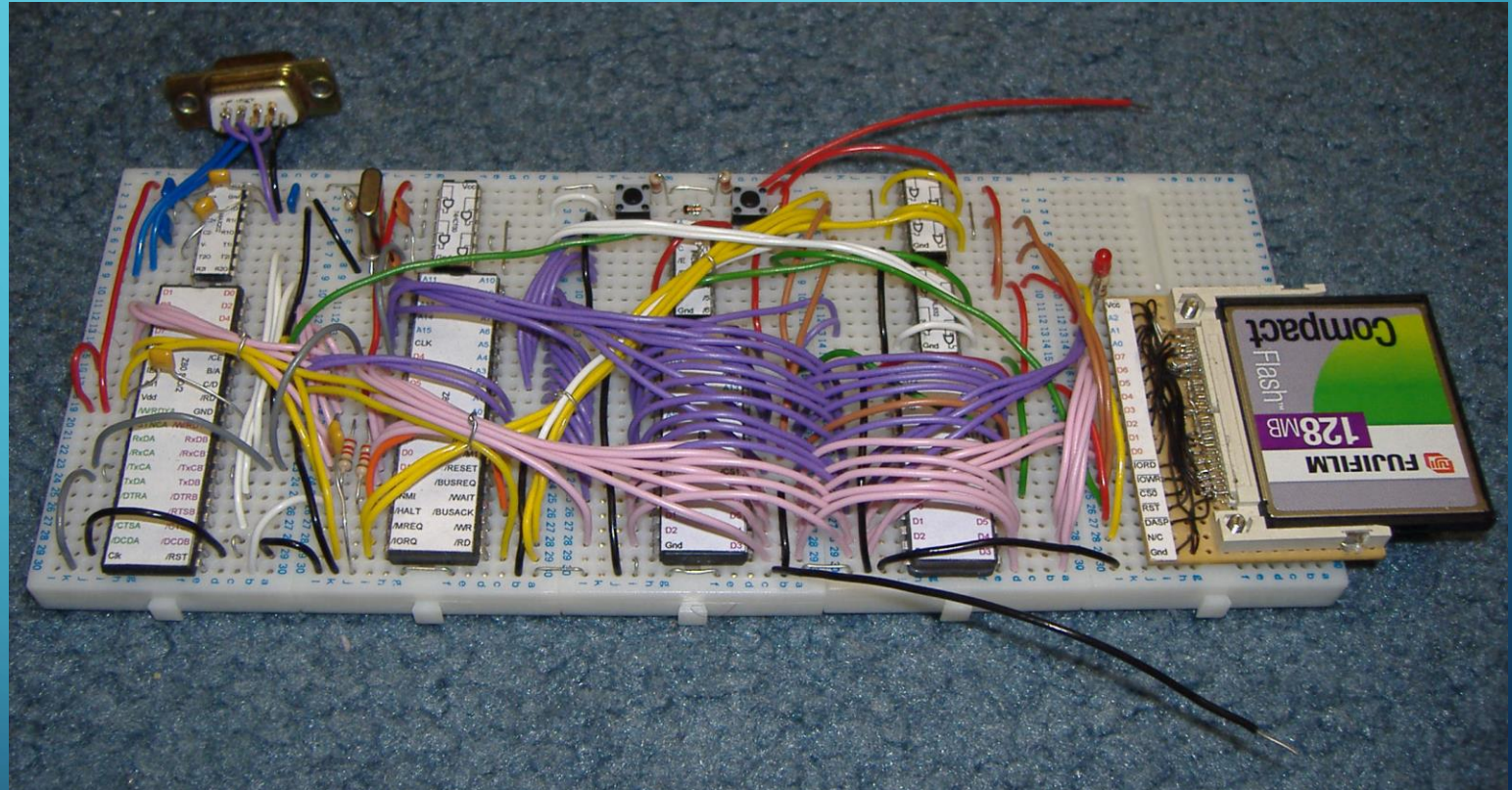
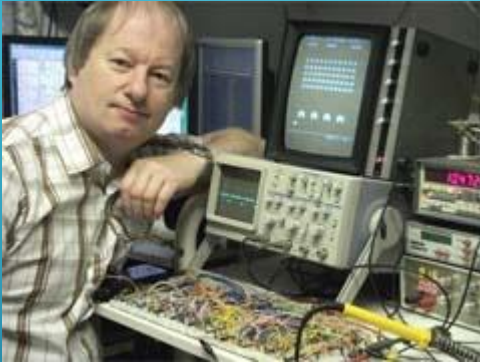
PROJECT INSPIRATION

RC2014



<https://www.youtube.com/watch?v=wVUIORo0-Ak>

WE ALL BUILD ON THE SHOULDERS OF GIANTS



GRANT HAS BASIC AND CP/M FIRMWARE

```
Z80 SBC By Grant Searle
```

```
Memory top?
```

```
Z80 BASIC Ver 4.7b
```

```
Copyright (C) 1978 by Microsoft
```

```
56958 Bytes free
```

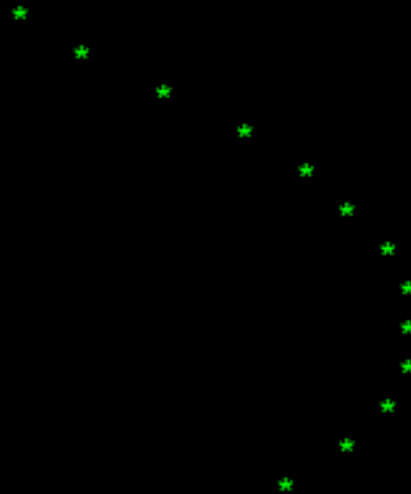
```
Ok
```

```
10 FOR A=0 TO 6.2 STEP 0.2
```

```
20 PRINT TAB(40+SIN(A)*20);"*"
```

```
30 NEXT A
```

```
RUN
```



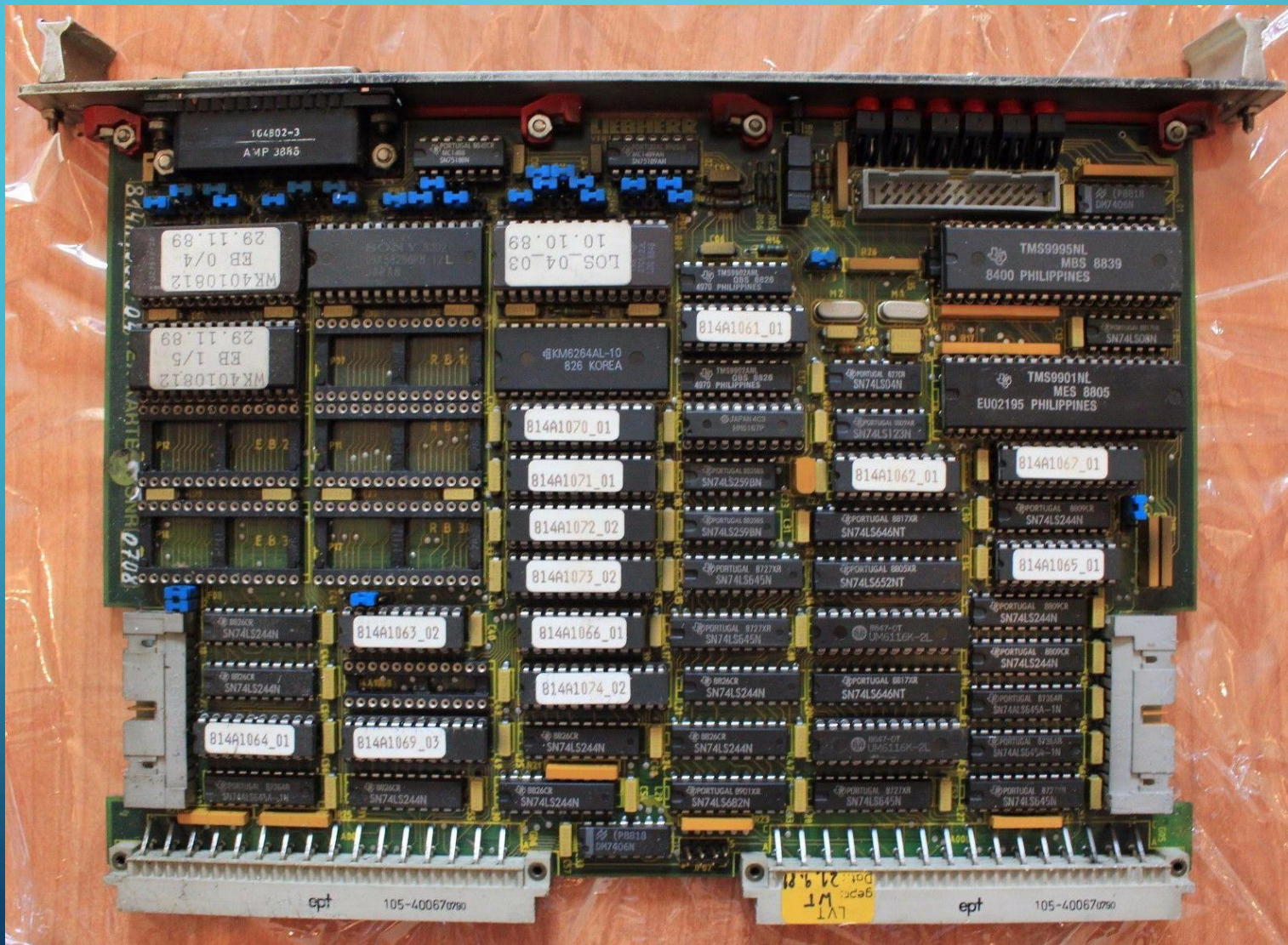
WHAT DO YOU DO WITH YOUR RC2014 AFTER CP/M?



1980S CHIP STASH



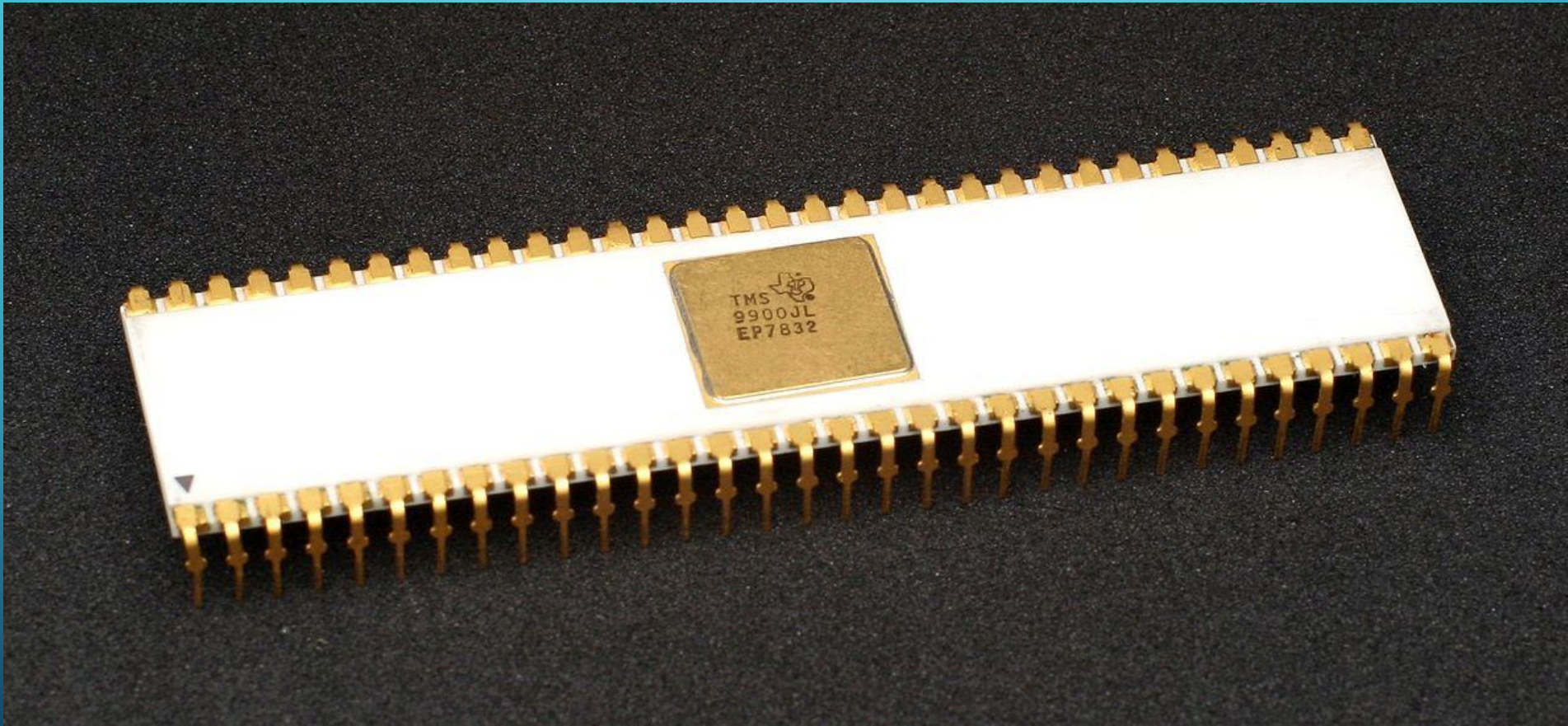
LIEBHERR 814A1001 PLC



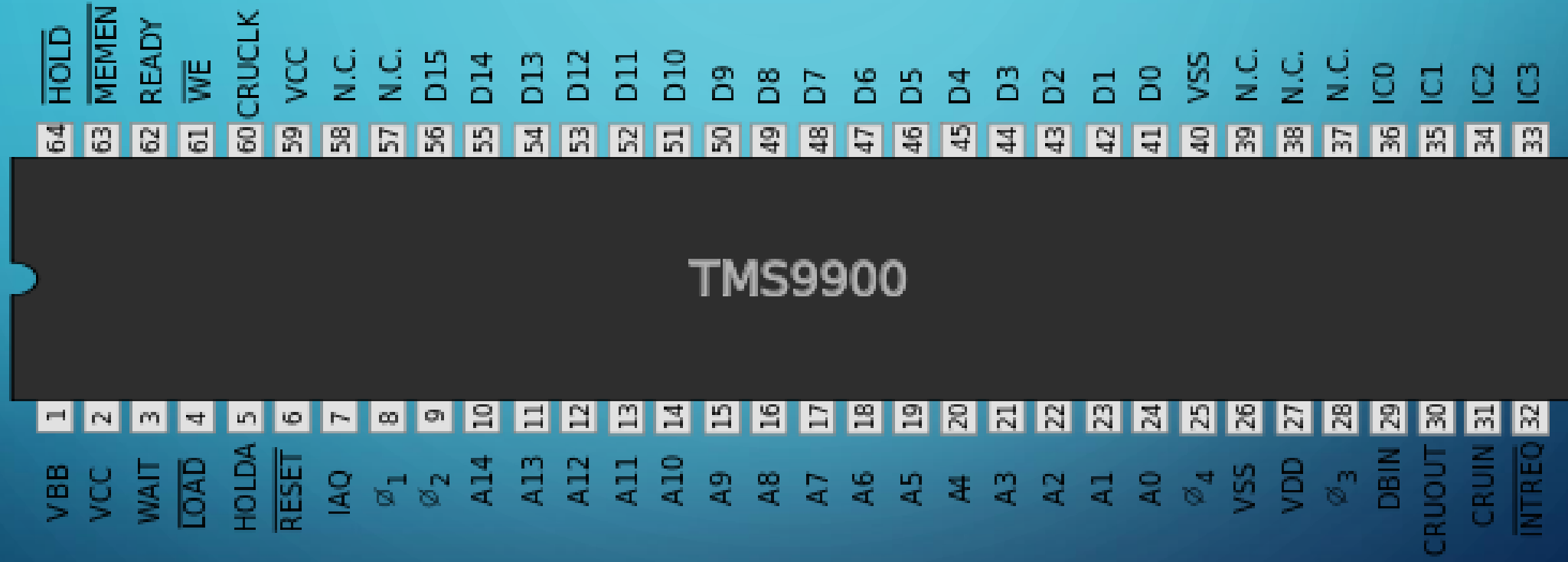
The background is a blue gradient with decorative white circuit-like lines in the corners. These lines consist of straight segments and small circles, resembling a stylized PCB or network diagram.

TMS9995 ORIGINS

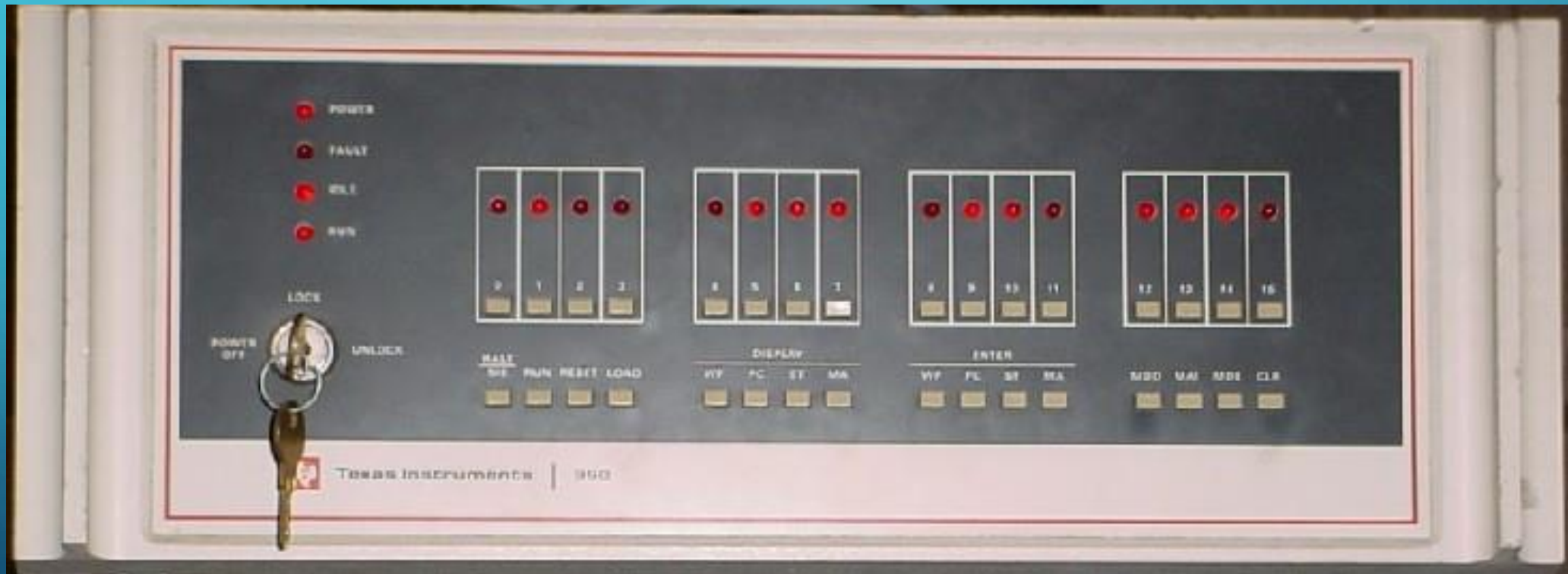
TMS9900 (1976)



TMS9900 PINOUT



TI-990 MINI COMPUTER (1973)



TI-99/4 (1979)



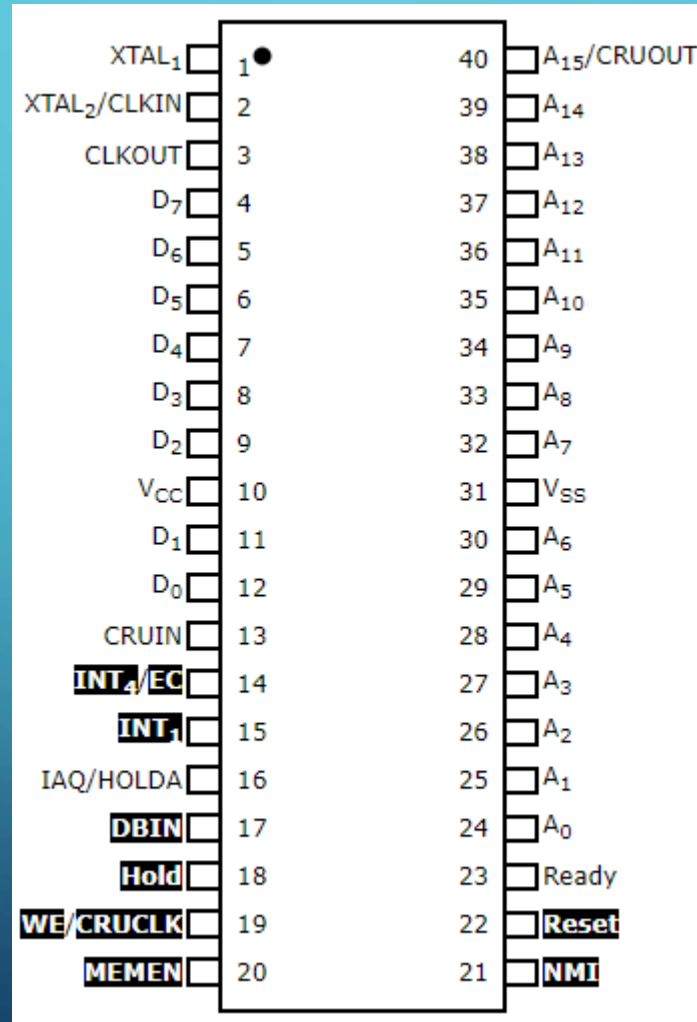
[https://commons.wikimedia.org/wiki/File:TI-99-4_homecomputer_\(adjusted_version\).jpg](https://commons.wikimedia.org/wiki/File:TI-99-4_homecomputer_(adjusted_version).jpg) CC ASA 4.0 by Tocchet22

TI-99/4A (1981)



https://commons.wikimedia.org/wiki/File:TI99-IMG_7132.jpg CC ASA 2.0 by Rama & Musée Bolo

TMS9995 (1982)



TI99/2 (1983)



<https://99er.net/992.html> Photo by Bryan Roppolo

TI99/8 (1983)



https://www.ninerpedia.org/wiki/File:99_8.jpg

TOMY TUTOR (1982)



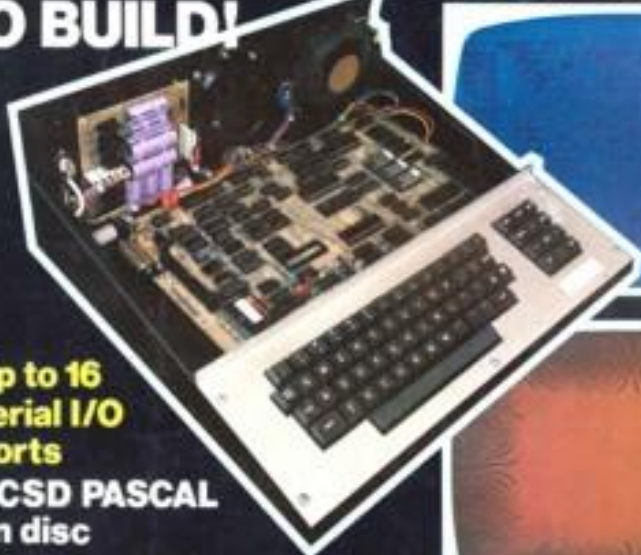
<https://commons.wikimedia.org/wiki/File:Tomy-Tutor-wControllers.jpg> Public Domain by Evan-Amos

POWERTRAN CORTEX

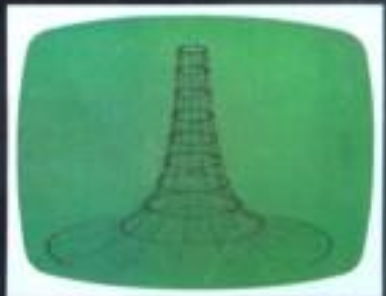
electronics today ISSN 0142-7228
INTERNATIONAL NOVEMBER 1982 85p

THE BIG ONE!

16-BIT COLOUR COMPUTER TO BUILD!



- Up to 16 serial I/O ports
- UCSD PASCAL on disc
- High resolution graphics
- Powerful resident BASIC
- Separate 16K video RAM
- Addressing up to 1M
- Up to 60K of user memory
- Floppy controller option
- Programmable baud rate

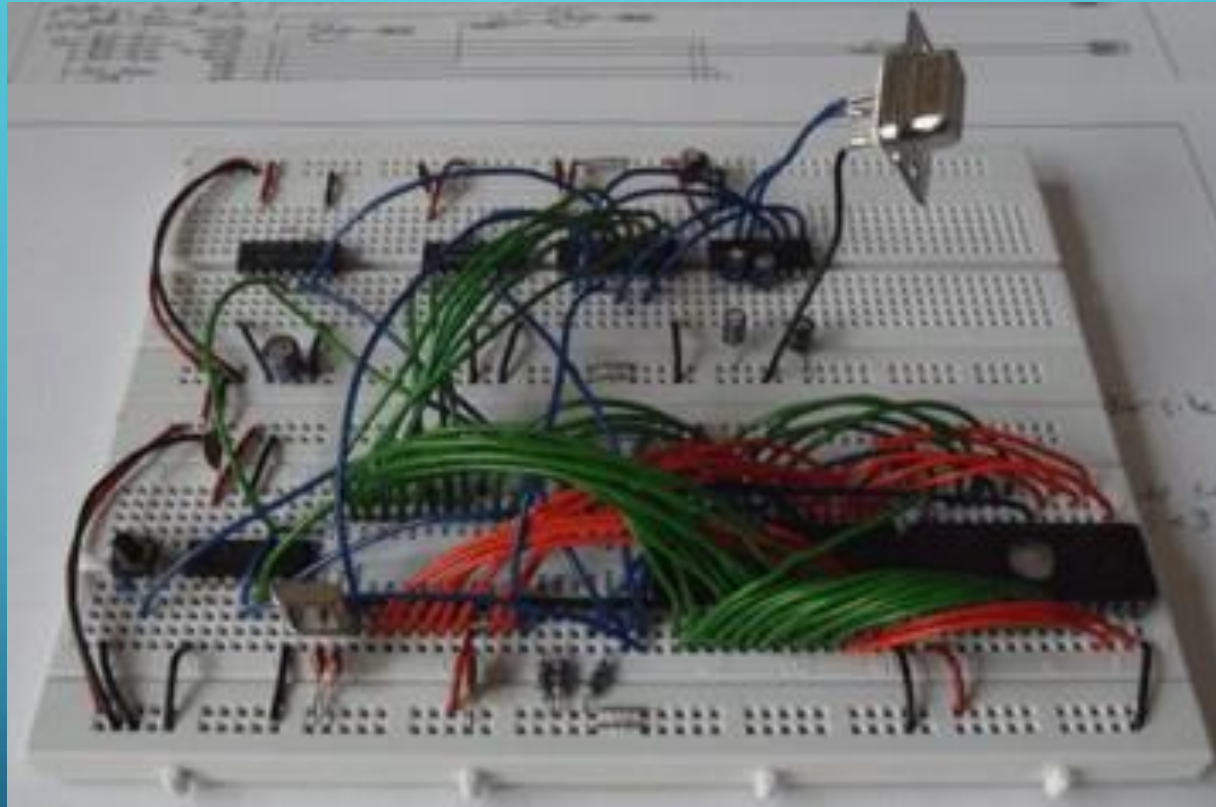


...AUDIO....COMPUTING....MUSIC....RADIO....ROBOTICS...



A DESIGN TO WORK FROM

STUART CONNER – THE GRANT SEARLE OF TMS 99XX



http://www.stuartconner.me.uk/tms9995_breadboard/tms9995_breadboard.htm

FIRMWARE

EPROM Binary Images

Link to EVMBUG system monitor EPROM image [here](#).

Link to EVMBUG system monitor and BASIC EPROM image [here](#) (BASIC runs from RAM).

Link to EVMBUG system monitor and BASIC EPROM image [here](#) (BASIC runs from EPROM).

Link to EVMBUG system monitor and Forth EPROM image [here](#).

Link to test EPROM image #1 [here](#) - see Testing and Fault-Finding section below.

Link to test EPROM image #2 [here](#) - see Testing and Fault-Finding section below.

Link to test EPROM image #3 [here](#) - see Testing and Fault-Finding section below.

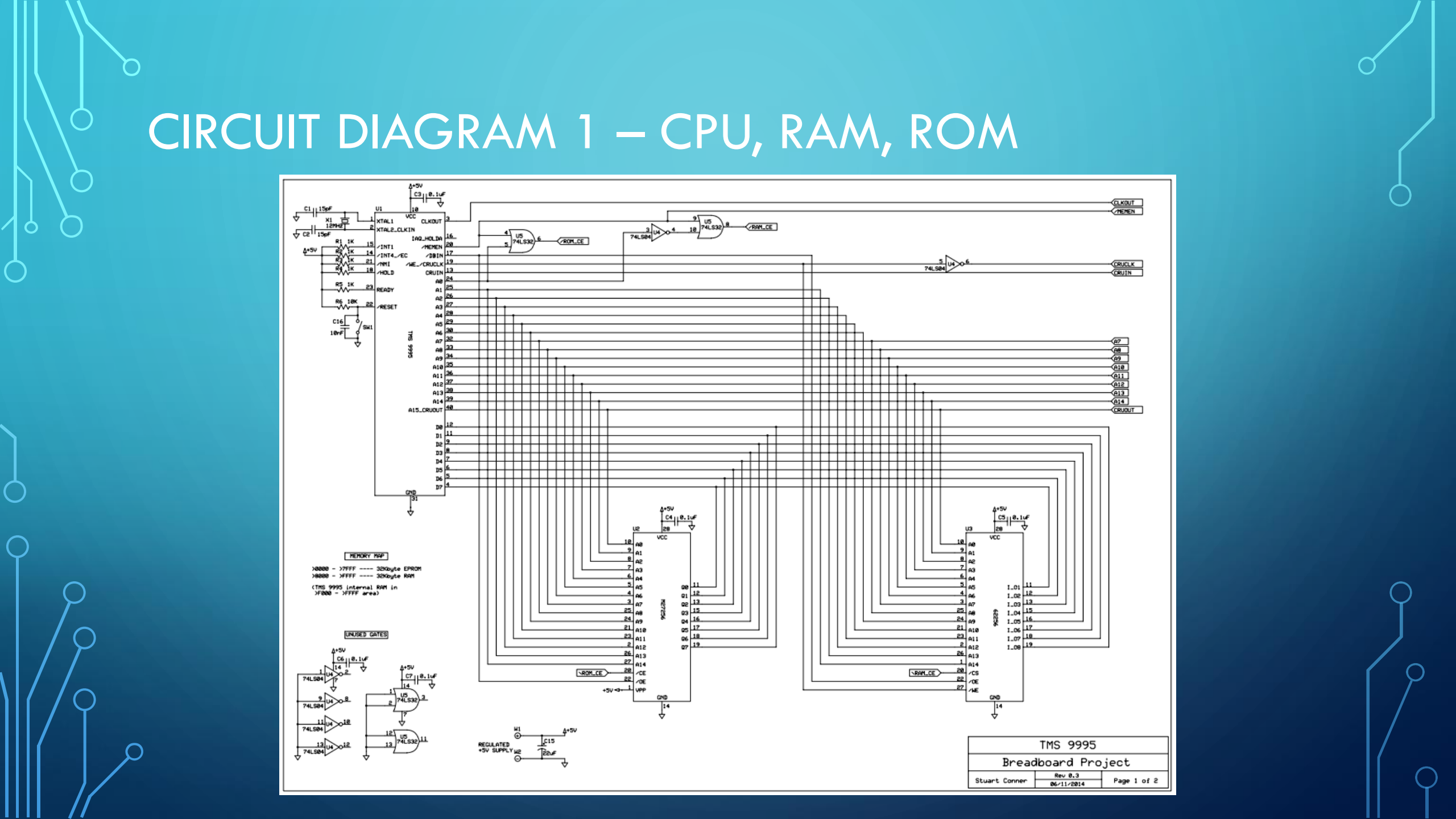
CIRCUIT DIAGRAM 1 – CPU, RAM, ROM

The diagram illustrates the internal components and connections of the TMS9995 microcontroller. It includes the following sections:

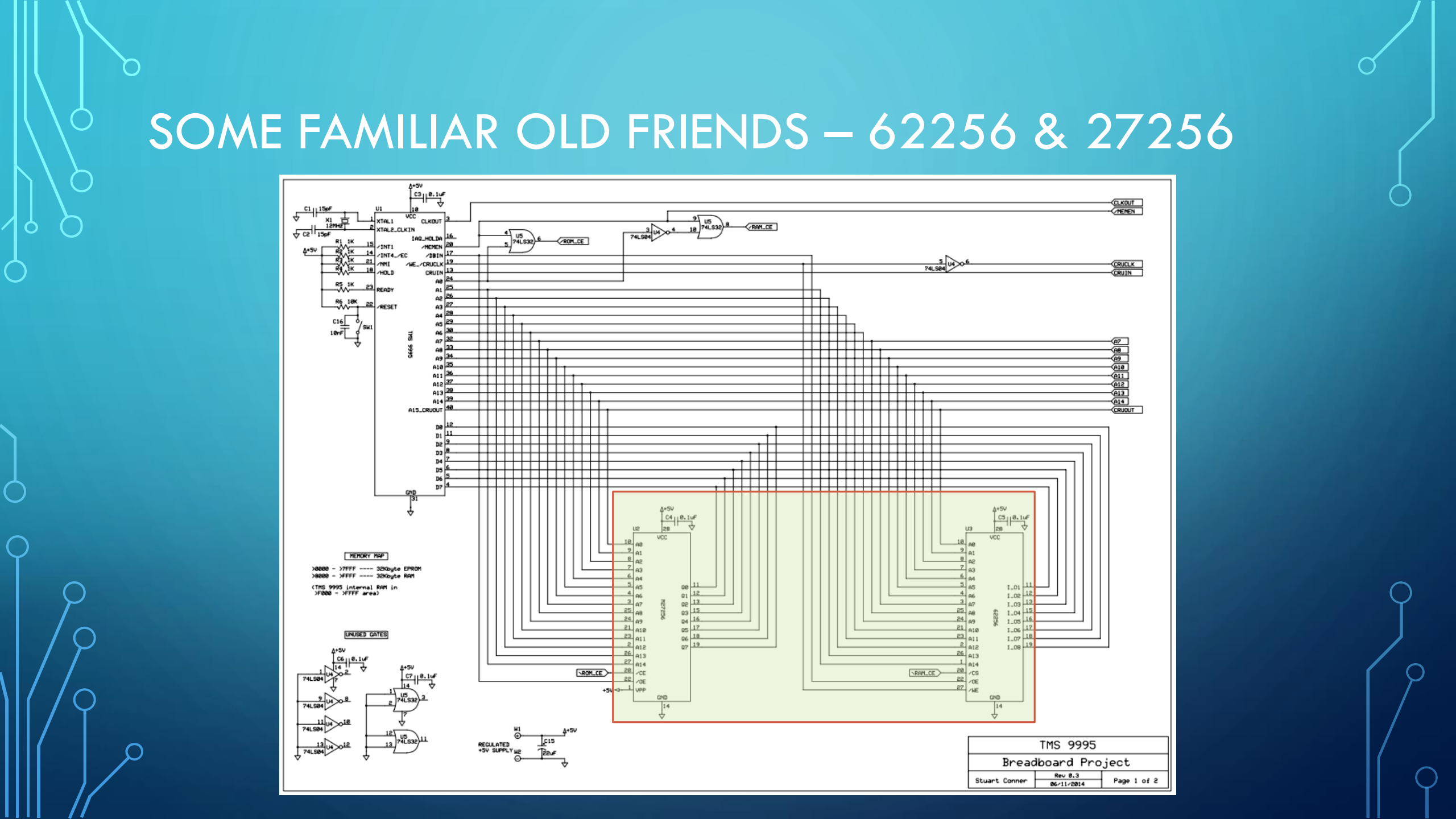
- Memory Map:**
 - >0000 - >7FFF ---- 32Kbyte EPROM
 - >8000 - >FFFF ---- 32Kbyte RAM
 - (TMS 9995 internal RAM in >F000 - >FFFF area)
- Unused Gates:**
 - 74LS04 (Inverters)
 - 74LS32 (OR gate)
 - 74LS33 (Schmitt trigger)
- Regulated +5V Supply:**
 - W1: +5V
 - W2: GND
 - C15: 22μF
- Internal RAM and ROM:**
 - Internal RAM (A0-A14, D0-D7)
 - Internal ROM (A0-A14, D0-D7)
- Logic Chips:**
 - 74LS04 (Inverters)
 - 74LS32 (OR gate)
 - 74LS33 (Schmitt trigger)

TMS 9995 Breadboard Project

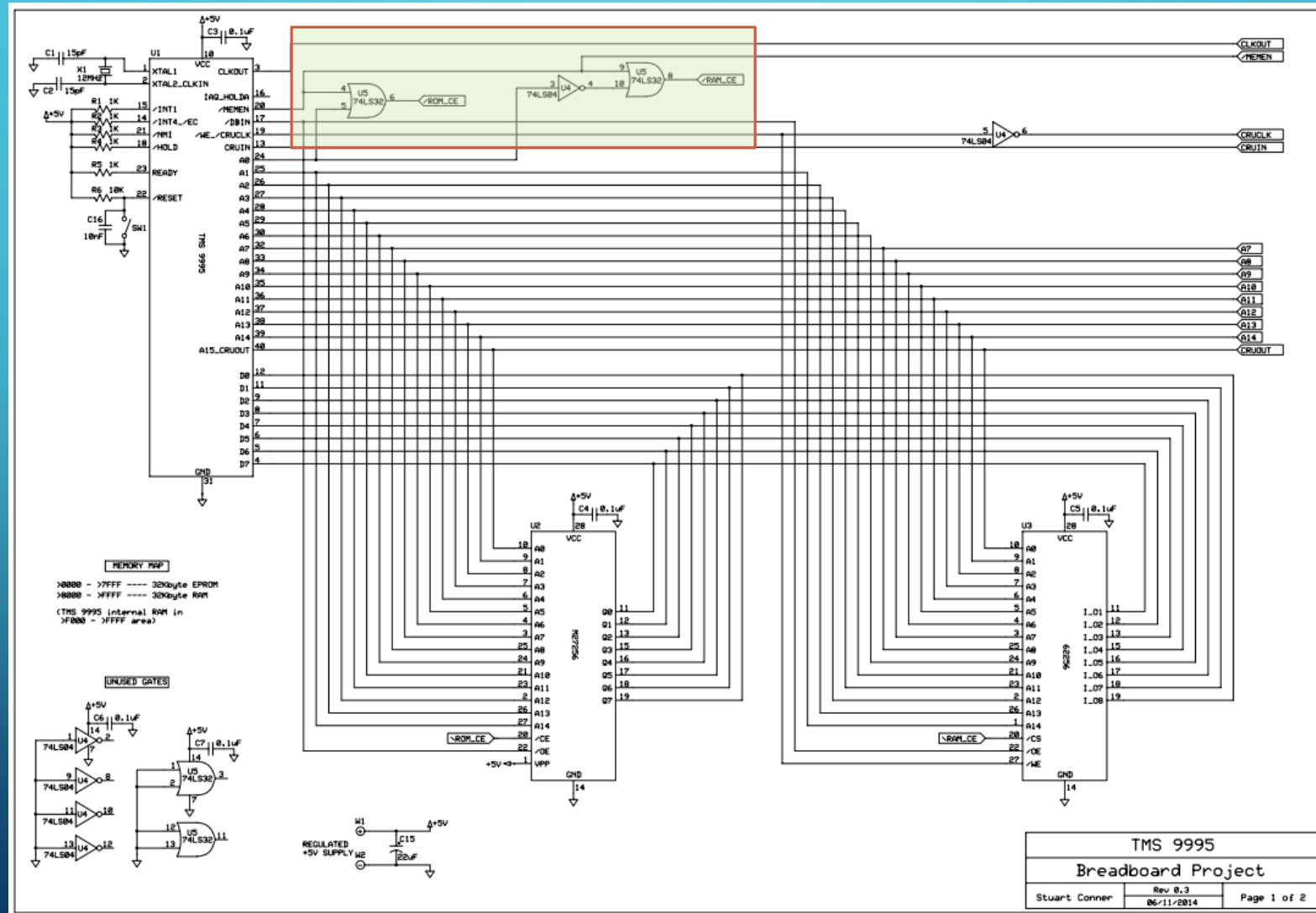
Stuart Conner Rev 0.3 06/11/2014 Page 1 of 2



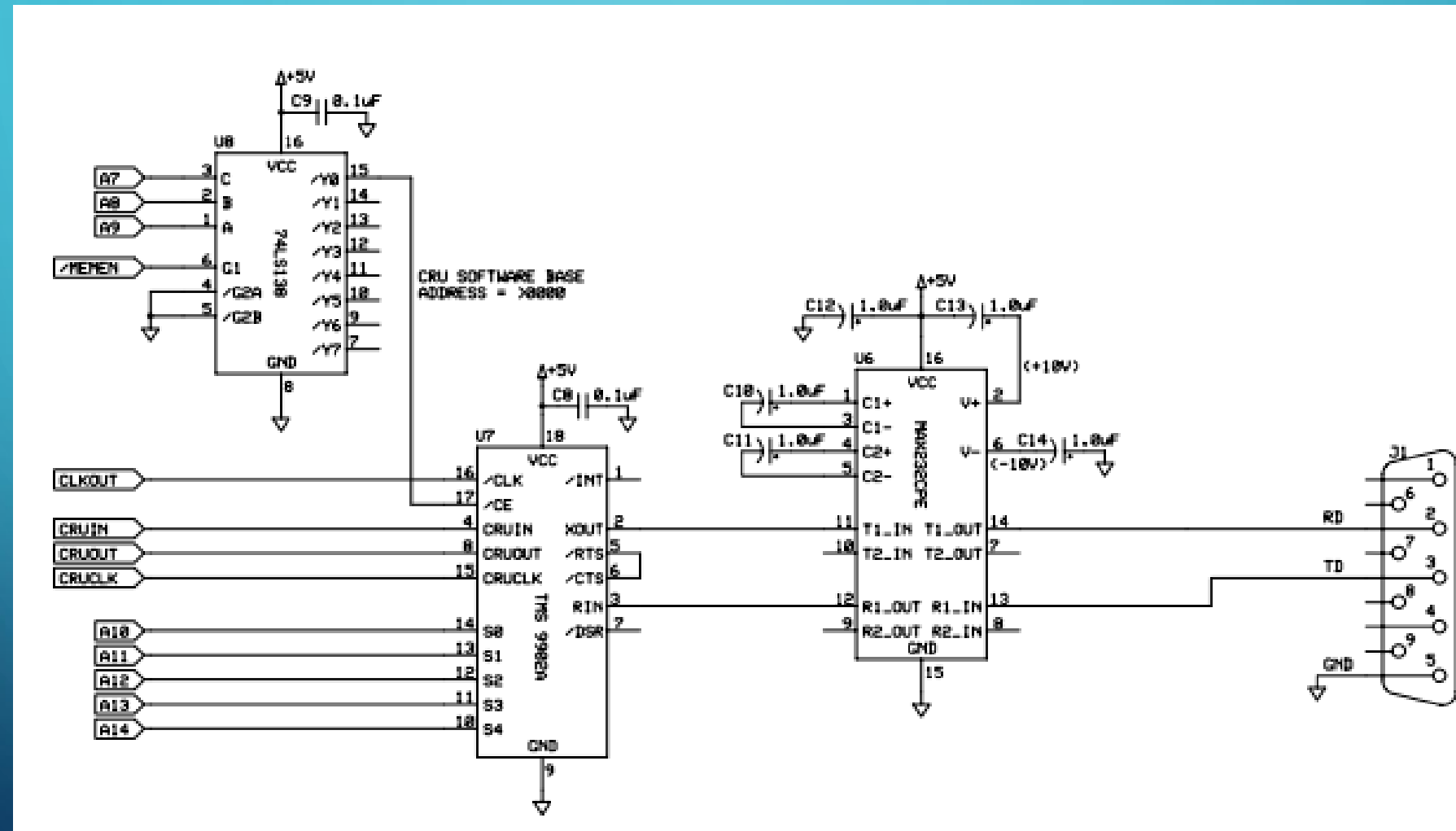
SOME FAMILIAR OLD FRIENDS – 62256 & 27256



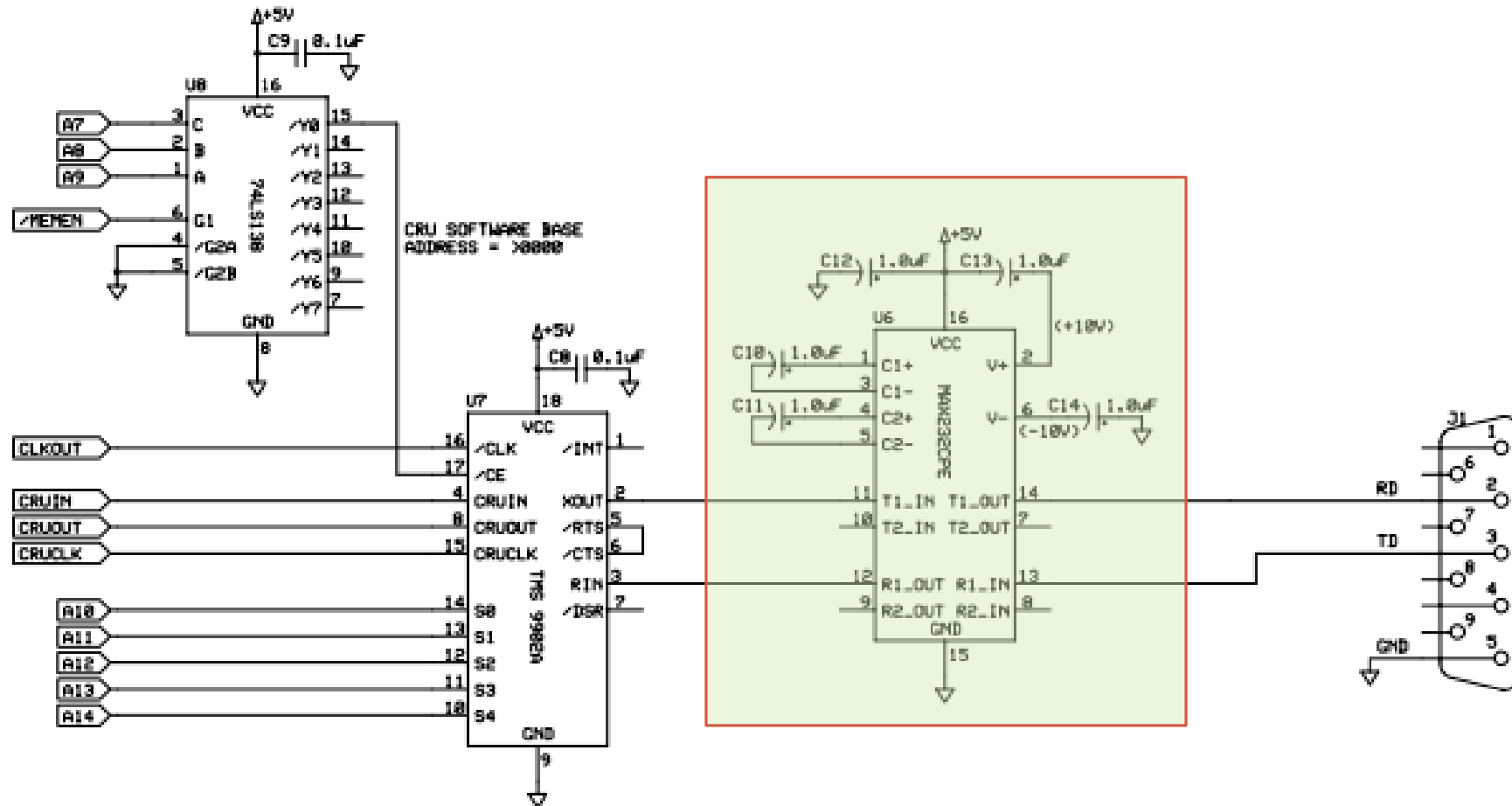
AND SOME SIMILAR CHIP SELECT LOGIC



SERIAL PORT IS A BIT DIFFERENT



BUT I DON'T ACTUALLY NEED RS232 LEVELS





PUTTING IT TOGETHER

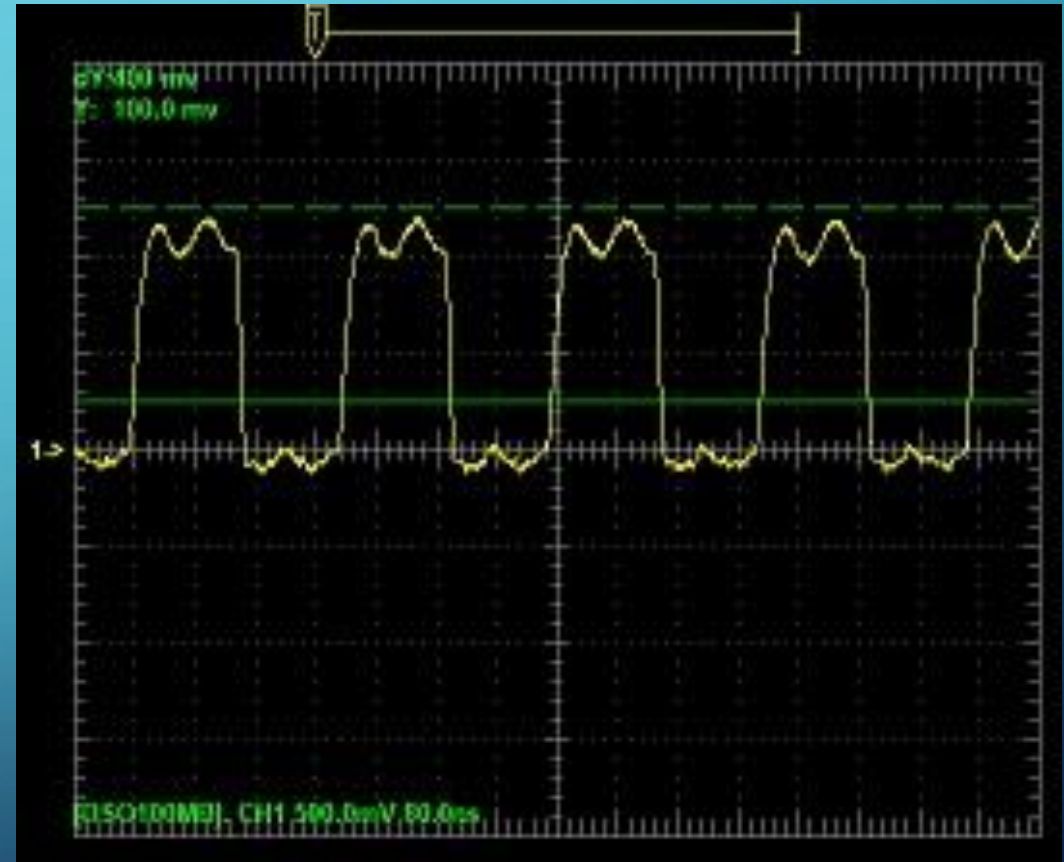
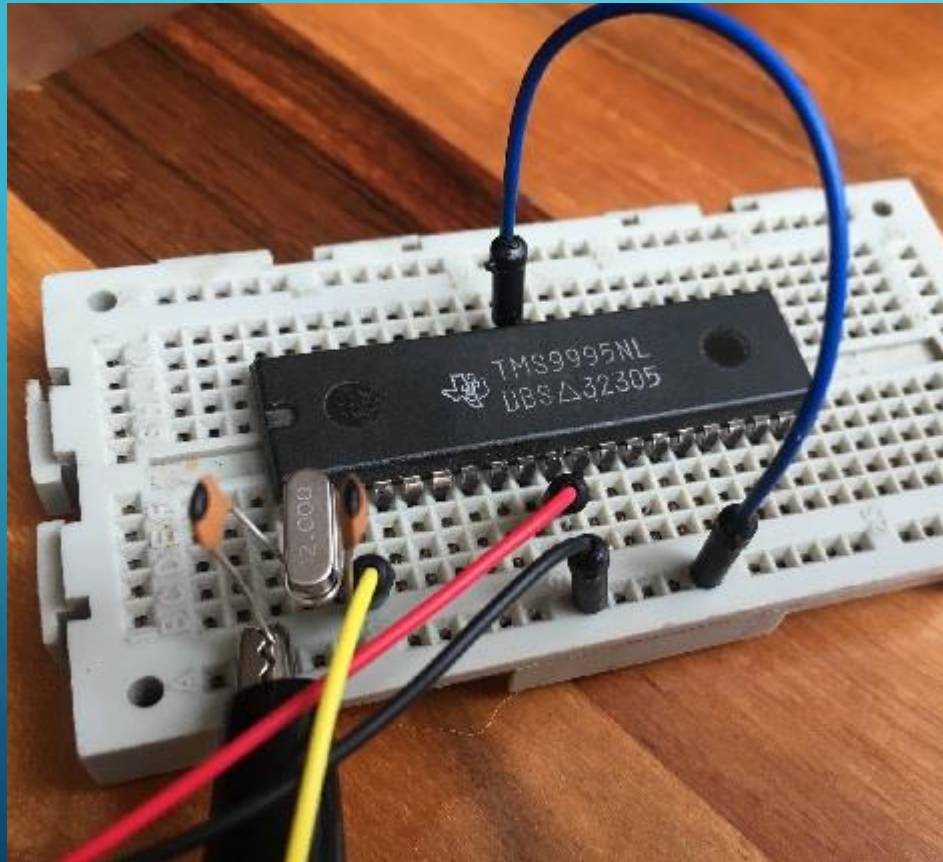
MAPPING ARCHITECTURES – FIRST THOUGHTS

Z80	TMS9995
MREQ	MEMEN
RD	DBIN
WR	WE

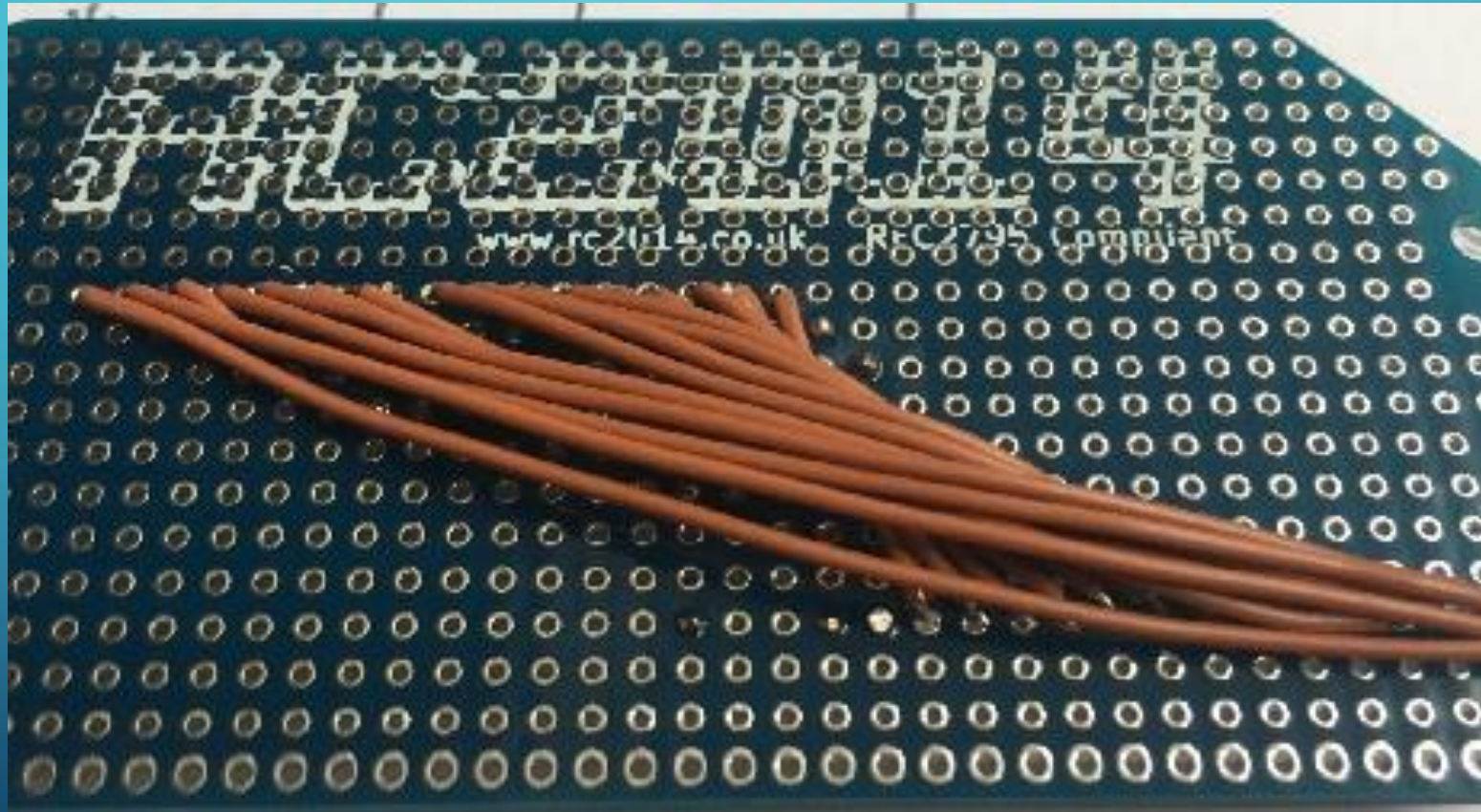
MAPPING ARCHITECTURES

TMS9995	RC2014
A0-A15	A15-A0
Vss	GND
Vcc	5v
NC	M1
Reset	Reset
CLKOUT	Clock
NC	INT
MEMEN	Mreq
WE/CRUCLK	Wr
DBIN	Rd
NC	IORQ
D7-D0	D0-D7
CRUIN	USR1

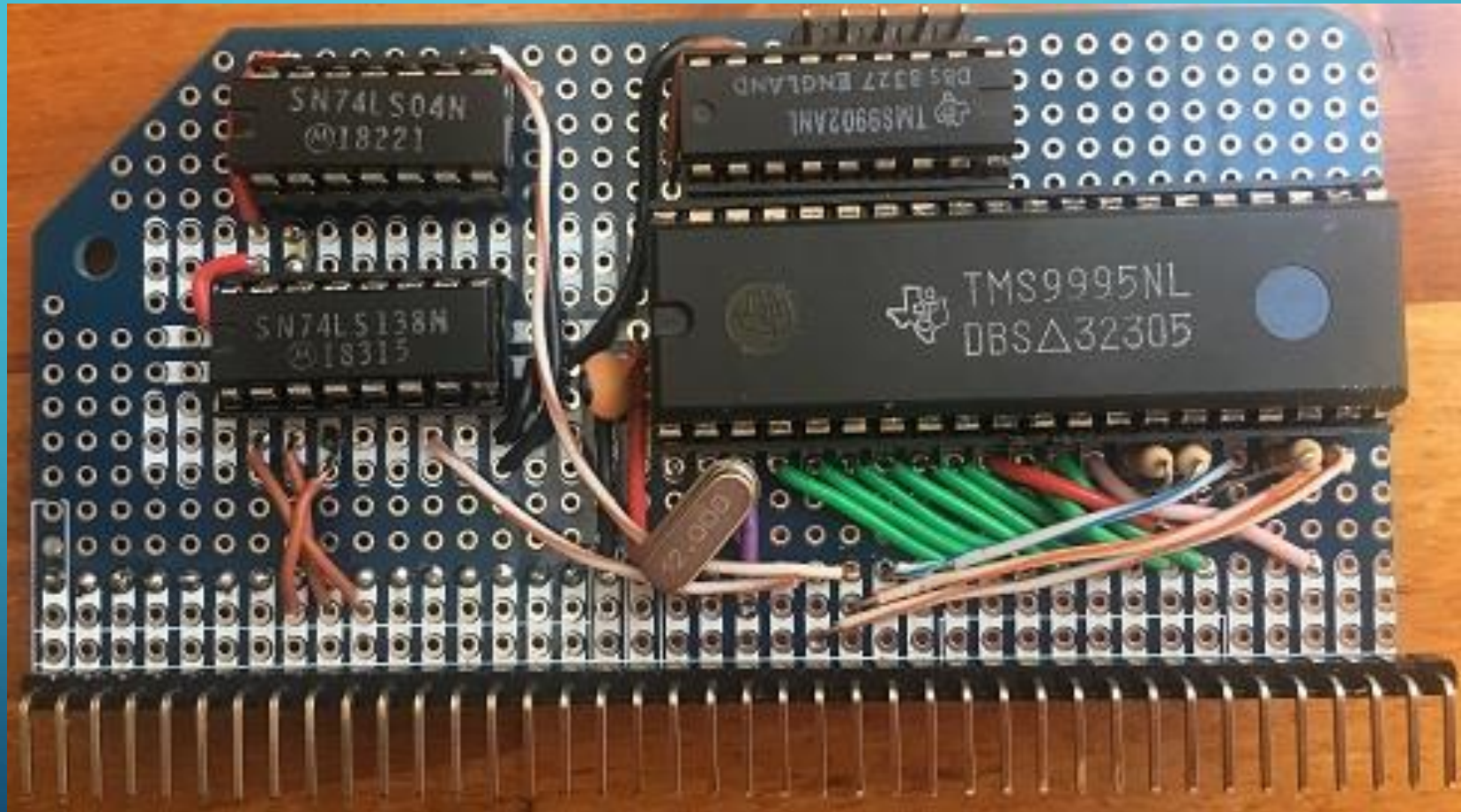
CLOCK CHECK



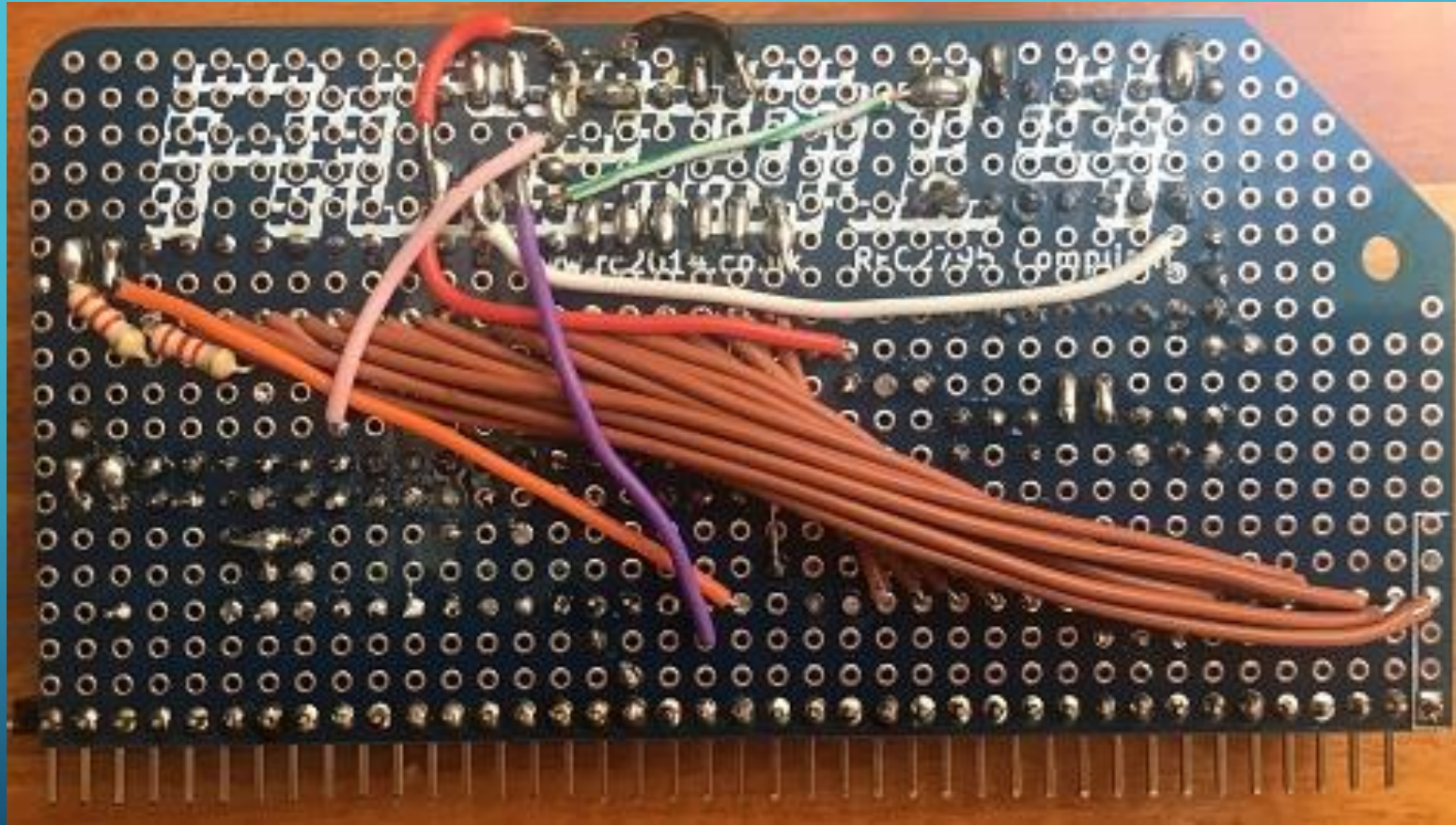
ADDRESS BUS



COMPLETED FRONT



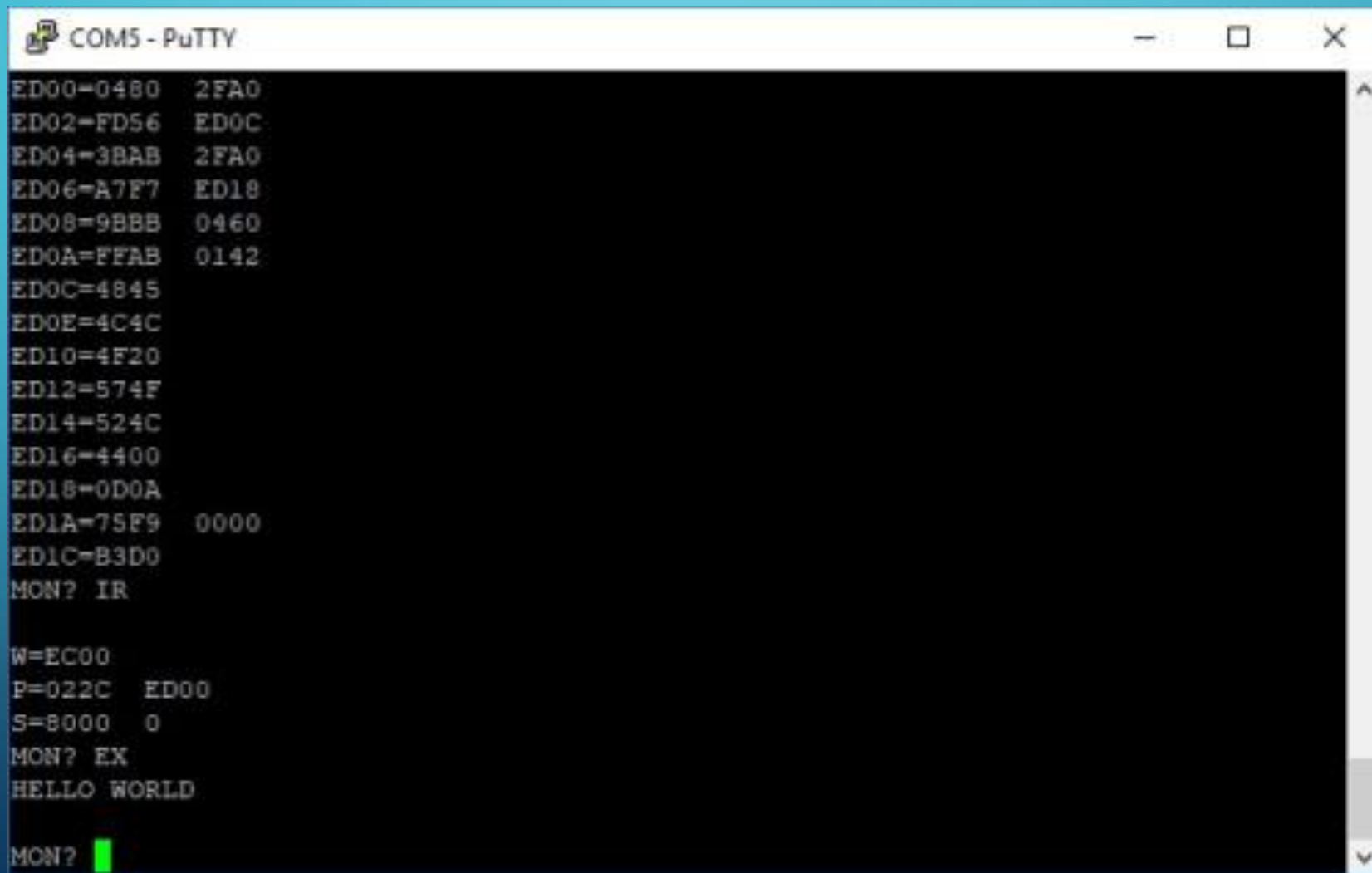
COMPLETED BACK



The background is a blue gradient with decorative circuit-like lines in the corners. The lines are white and light blue, forming a network of nodes and connections. The text "HELLO WORLD" is centered in the middle of the image.

HELLO WORLD

HELLO WORLD



```
COM5 - PuTTY
ED00=0480  2FA0
ED02=FD56  ED0C
ED04=3BAB  2FA0
ED06=A7F7  ED18
ED08=9BBB  0460
ED0A=FFAB  0142
ED0C=4845
ED0E=4C4C
ED10=4F20
ED12=574F
ED14=524C
ED16=4400
ED18=0D0A
ED1A=75F9  0000
ED1C=B3D0
MON? IR

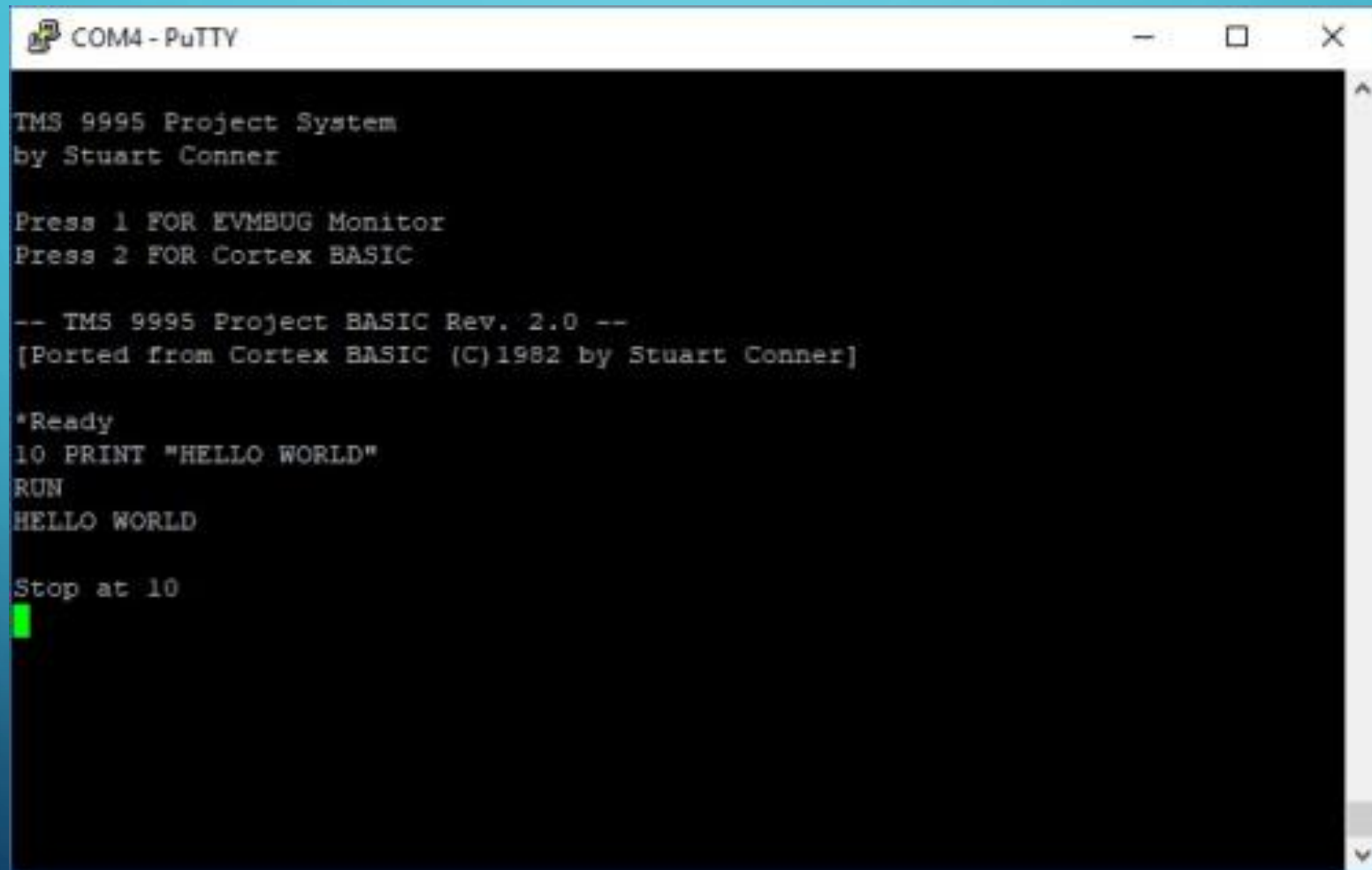
W=EC00
P=022C  ED00
S=8000  0
MON? EX
HELLO WORLD

MON? █
```

HACKING THE RC2014 ROM MODULE



BASIC



```
COM4 - PuTTY

TMS 9995 Project System
by Stuart Conner

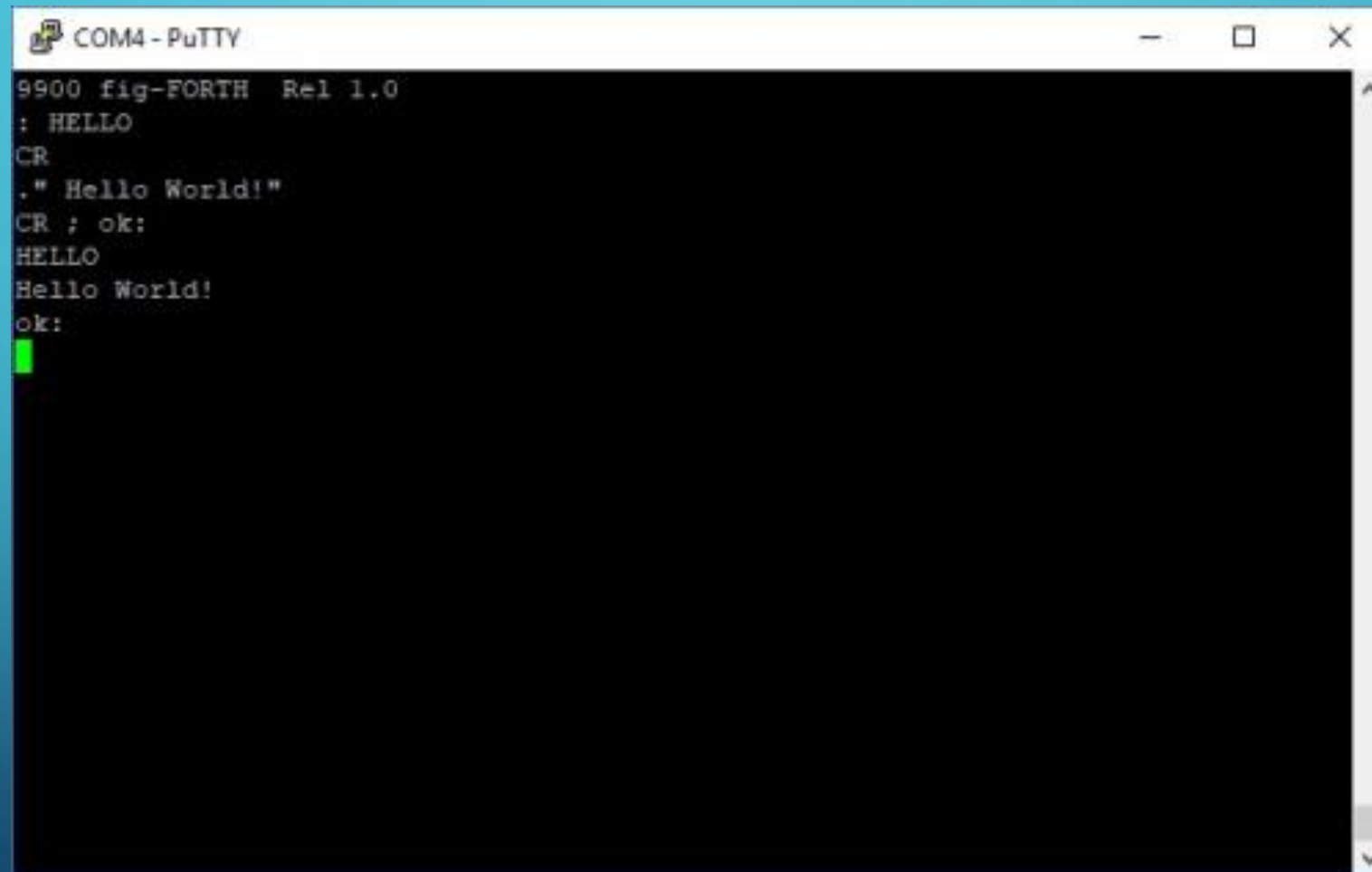
Press 1 FOR EVMBUG Monitor
Press 2 FOR Cortex BASIC

-- TMS 9995 Project BASIC Rev. 2.0 --
[Ported from Cortex BASIC (C)1982 by Stuart Conner]

*Ready
10 PRINT "HELLO WORLD"
RUN
HELLO WORLD

Stop at 10
```

FORTH



```
COM4 - PuTTY
9900 fig-FORTH Rel 1.0
: HELLO
CR
." Hello World!"
CR ; ok:
HELLO
Hello World!
ok:
█
```

The screenshot shows a terminal window titled "COM4 - PuTTY". The text inside the window is as follows: "9900 fig-FORTH Rel 1.0", ": HELLO", "CR", ".\" Hello World!\"", "CR ; ok:", "HELLO", "Hello World!", "ok:", and a green cursor. The window has standard Windows-style window controls (minimize, maximize, close) in the top right corner.

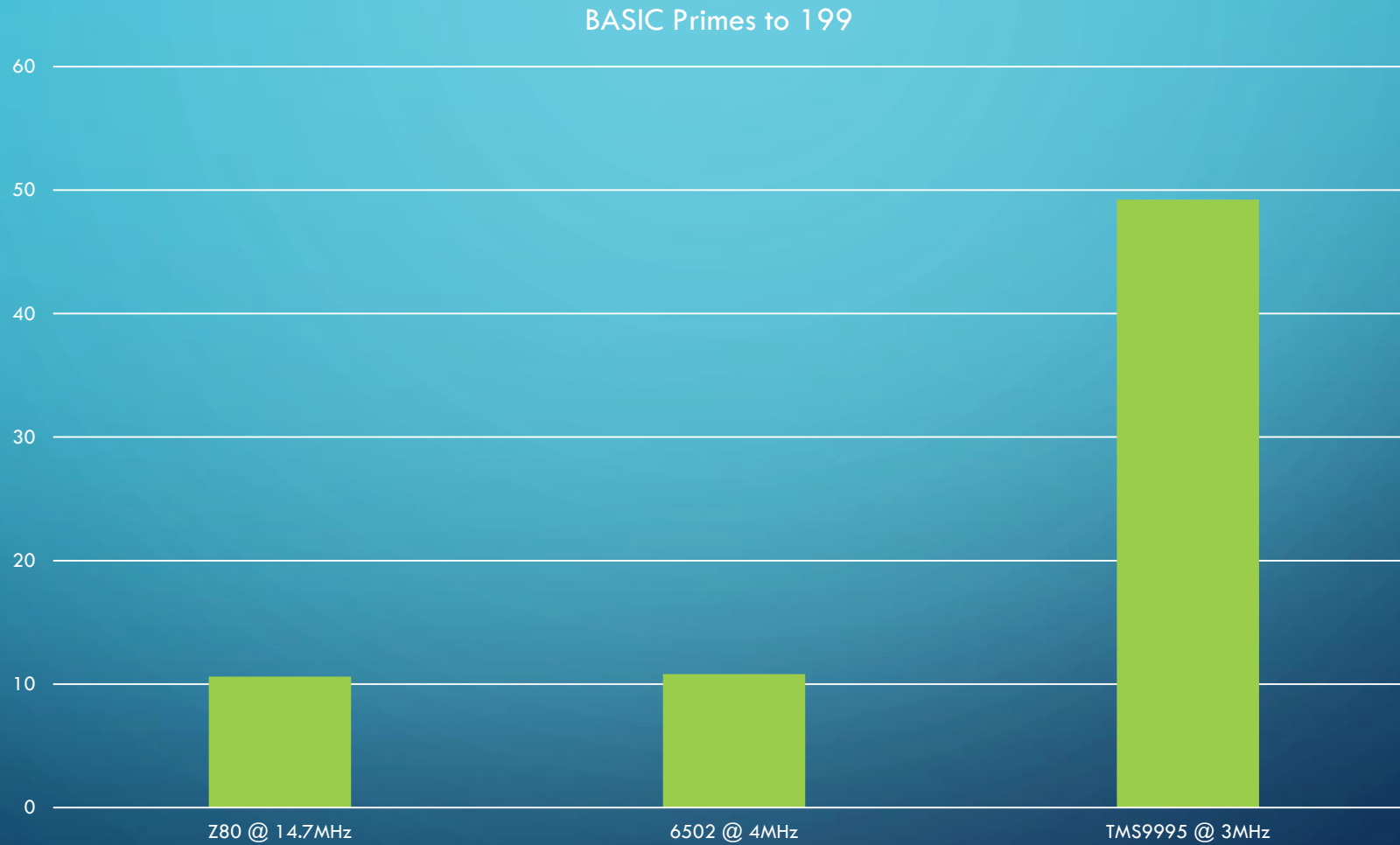
FINISHING TOUCHES



The background is a blue gradient with decorative white circuit-like lines in the corners. These lines consist of straight segments and small circles, resembling a stylized electronic circuit board.

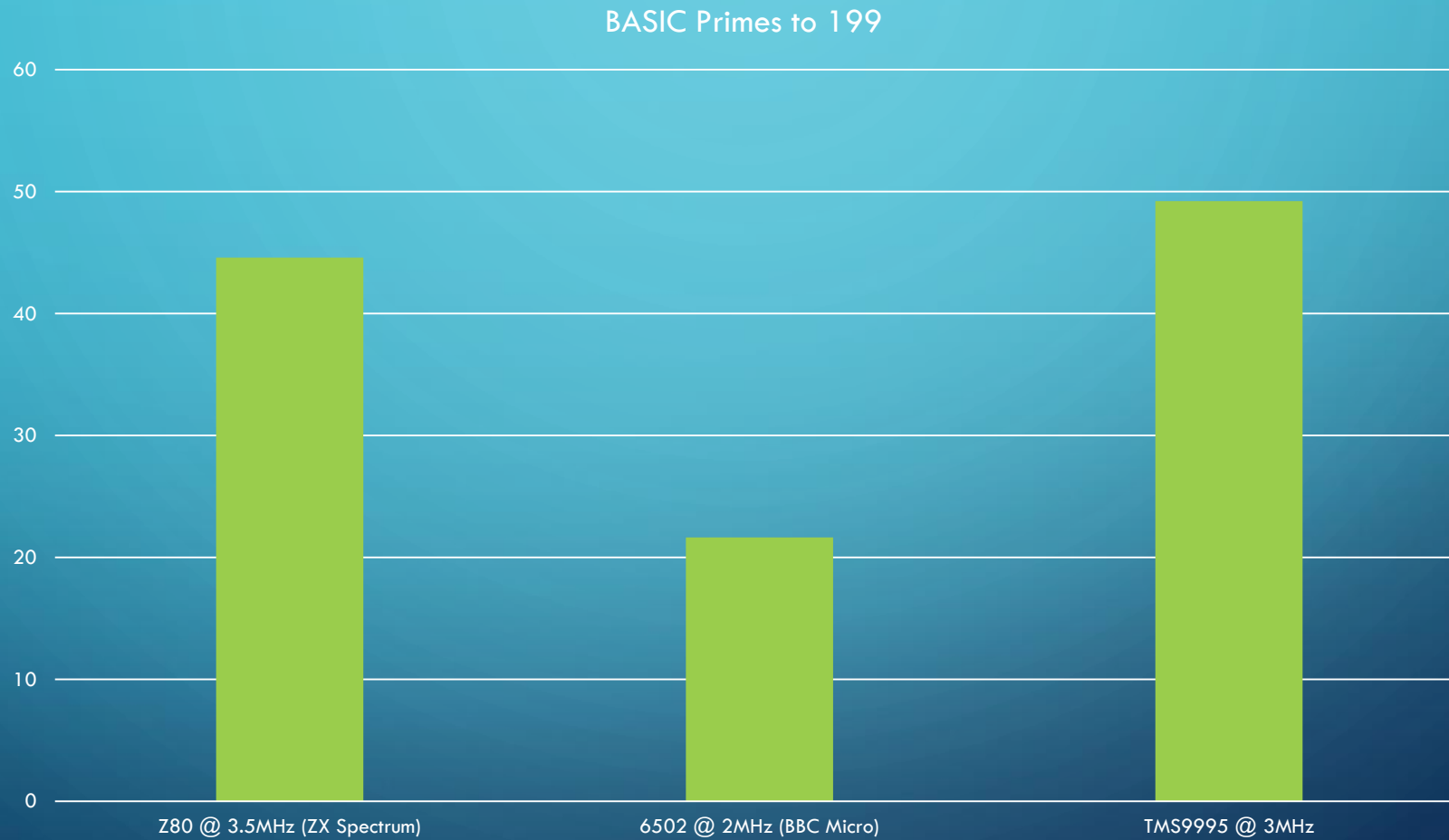
ARE 16BITS BETTER?

NO!



<http://ancientcomputing.blogspot.com/2017/06/a-6502-cpu-for-rc2014-software.html>

NOT GREAT, NOT TERRIBLE



TMS99XX (KIND OF) LIVES ON IN MSP430 MCUS

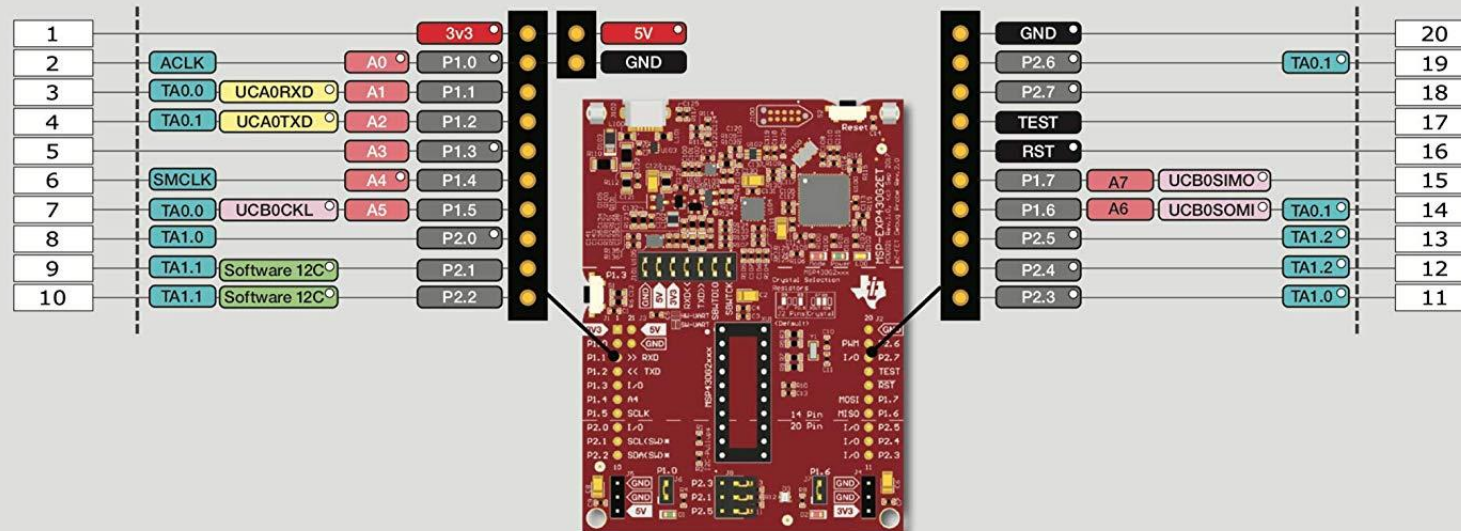


TI LaunchPad™ kit with MSP430™ MCU

Microcontroller development kit for rapid prototyping
featuring the MSP430G2553 microcontroller

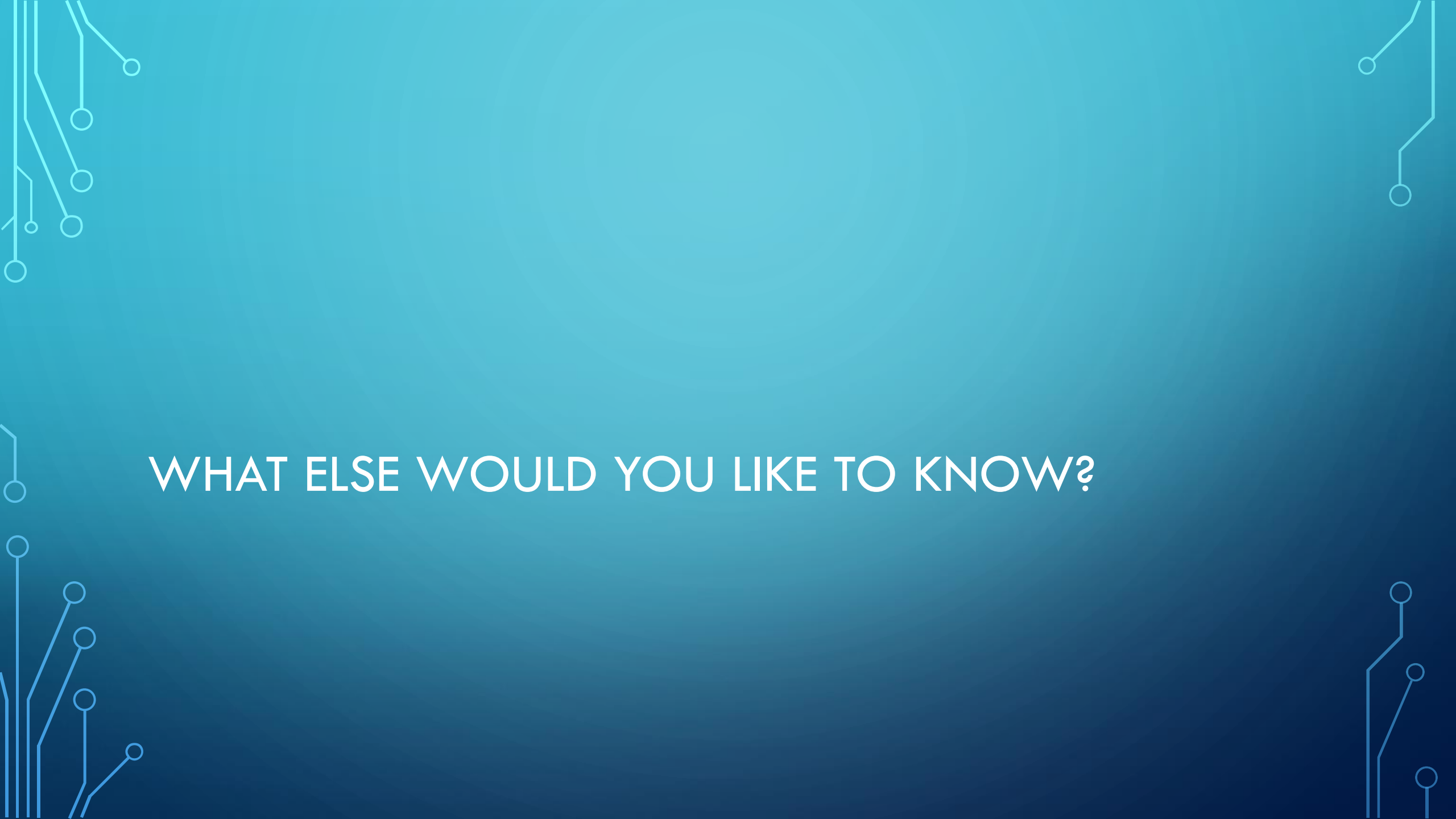
PART NO: MSP-EXP430G2ET

○ Pin aligns with LaunchPad pinout standard.



Explore all TI LaunchPad™ kits @ www.ti.com/launchpad

© 2018 Texas Instruments Incorporated. The platform bar, MSP430 and LaunchPad are trademarks of Texas Instruments.
All other trademarks are the property of their respective owners.

The background is a blue gradient with abstract white lines and circles in the corners, resembling a circuit or network diagram.

WHAT ELSE WOULD YOU LIKE TO KNOW?